

SANTA BARBARA, CA, USA

WMO: 723925

Lat: 34.426N

Lon: 119.843W

Elev: 9

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23190

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	35.4	37.3	21.5	16.1	55.7	25.4	19.4	53.9	20.8	58.5	18.7	58.3	0.7	60	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.6	83.5	63.5	80.1	63.4	77.2	63.0	68.8	77.3	67.3	75.4	66.1	73.8	8.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	94.6	72.7	64.0	89.4	70.7	63.1	86.5	69.5	32.9	77.1	31.7	75.3	30.8	73.7	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.8	16.1	12.9	DB	30.9	92.8	2.4	4.3	29.1	96.0	27.7	98.5	26.4	100.9	24.6	104.1
			WB	28.5	70.5	2.4	1.9	26.8	71.9	25.4	73.0	24.0	74.1	22.2	75.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.6	53.3	54.0	56.2	58.0	60.4	62.7	65.8	66.3	65.7	62.4	57.3	52.8
	DBStd	6.23	4.29	4.27	4.15	4.42	3.97	3.06	3.13	3.14	4.03	4.40	4.40	4.14
	HDD50	51	16	10	3	1	0	0	0	0	0	0	3	18
	HDD65	2205	363	308	274	217	156	82	27	20	37	107	235	379
	CDD50	3555	118	123	195	241	322	381	488	505	472	384	221	105
	CDD65	231	0	0	2	7	12	13	50	60	58	26	3	0
	CDH74	1509	22	29	46	89	112	68	144	233	339	324	97	6
	CDH80	320	1	2	6	22	33	13	20	30	73	98	22	0
(14) Wind	WSAvg	5.2	4.1	5.2	5.7	6.5	6.3	5.8	5.7	5.3	5.0	4.5	4.1	4.1
(15) Precipitation	PrecAvg	16.0	3.2	3.7	3.1	0.9	0.2	0.0	0.0	0.1	0.1	0.5	2.0	2.2
	PrecMax	35.1	12.2	13.6	11.5	5.7	1.8	0.5	0.4	1.9	1.3	2.4	6.9	5.8
	PrecMin	4.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	6.7	3.1	3.3	2.8	1.4	0.4	0.1	0.1	0.3	0.3	0.6	2.0	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.0	78.5	79.6	82.7	84.6	81.5	82.2	83.8	86.6	89.3	82.8	74.5
		MCWB	54.7	57.1	59.2	59.9	60.6	63.6	67.0	68.3	65.3	61.6	58.6	55.5
	2%	DB	71.7	71.9	73.1	75.4	76.2	75.0	77.4	79.3	81.4	81.9	76.1	70.0
		MCWB	54.2	55.9	57.0	57.5	60.1	62.9	66.2	67.0	66.8	61.0	57.5	54.6
	5%	DB	67.4	66.8	69.1	71.2	71.9	72.3	74.9	76.3	77.6	76.9	71.7	66.3
		MCWB	54.2	54.5	55.9	57.1	59.4	62.1	65.4	66.0	65.5	60.8	57.3	54.4
	10%	DB	63.8	63.3	65.4	67.4	68.4	70.0	72.7	73.7	74.6	72.9	68.0	63.4
		MCWB	54.8	53.9	55.2	56.3	58.7	61.6	64.5	65.0	64.8	61.0	57.0	53.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.7	61.0	62.2	63.3	65.3	66.3	69.7	69.9	71.0	69.3	63.6	61.5
		MCDB	65.8	68.6	72.5	75.0	75.6	75.7	77.4	80.0	79.3	77.5	72.5	66.5
	2%	WB	59.3	59.0	59.9	60.5	62.8	64.6	67.5	68.1	69.0	66.0	61.7	59.3
		MCDB	63.6	64.8	67.2	69.7	71.3	72.3	74.9	77.1	77.1	74.1	68.6	64.1
	5%	WB	57.7	57.5	58.5	59.2	61.3	63.5	66.2	66.7	67.2	64.1	60.3	57.5
		MCDB	62.2	62.8	64.8	67.1	68.7	70.3	73.5	74.8	74.8	71.3	66.9	62.1
	10%	WB	56.3	56.2	57.3	57.8	59.9	62.3	64.9	65.6	65.8	62.5	59.0	55.8
		MCDB	61.2	61.4	63.0	65.0	66.3	68.5	71.9	72.9	72.9	69.3	65.5	61.1
(35) Mean Daily Temperature Range	5% DB	MDBR	21.5	19.5	18.0	18.3	16.4	14.8	14.7	15.6	17.5	20.7	22.4	22.0
		MCDBR	29.3	27.5	25.8	25.8	23.0	19.9	18.0	19.9	23.4	29.4	29.3	28.1
		MCWBR	16.4	14.9	13.4	12.8	11.3	10.3	8.8	9.6	11.3	14.6	15.8	16.8
	5% WB	MCDBR	19.7	19.5	19.4	20.9	18.5	17.0	16.4	18.1	19.7	21.6	22.2	20.8
		MCWBR	13.8	12.0	11.7	11.7	10.1	9.2	8.1	8.9	9.9	12.4	14.0	14.5
(40) Clear-Sky Solar Irradiance	taub		0.304	0.318	0.326	0.364	0.379	0.365	0.395	0.374	0.363	0.351	0.309	0.311
	taud		2.592	2.566	2.543	2.421	2.396	2.438	2.409	2.435	2.474	2.500	2.599	2.585
	Ebn at Noon		290	297	302	292	287	290	280	286	285	280	286	280
	Edh at Noon		24	28	31	36	38	36	37	36	33	30	24	23
(44) All-Sky Solar Radiation	RadAvg		974	1248	1685	2102	2320	2411	2332	2184	1869	1433	1099	882
	RadStd		74	131	109	119	152	111	92	82	75	105	60	71

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)		+1.10	N/A	-2.92	+1.22	N/A	N/A	N/A	-229	+396	+149

Nomenclature: See separate page